

# Special Functions

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## Problem S.1

Determine the values of  $\Gamma(3)$ ,  $\Gamma(\frac{1}{2})$ ,  $\Gamma(-\frac{1}{2})$ .

### Solution:

Using the definition of  $\Gamma(z)$ :

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt$$

it can be shown that  $\Gamma(z+1) = z\Gamma(z)$ :

$$\begin{aligned} \Gamma(z+1) &= \int_0^{\infty} t^z e^{-t} dt = \underbrace{[-t^z e^{-t}]_0^{\infty}}_{=0} + \int_0^{\infty} z t^{z-1} e^{-t} dt = \\ &= z \int_0^{\infty} t^{z-1} e^{-t} dt = z\Gamma(z) \end{aligned}$$

We also have

$$\Gamma(1) = \int_0^{\infty} e^{-t} dt = [-e^{-t}]_0^{\infty} = 1$$

This gives

$$\Gamma(3) = \Gamma(2+1) = 2\Gamma(2) = 2\Gamma(1+1) = 2 \cdot 1 \cdot \Gamma(1) \rightarrow \Gamma(3) = 2$$

Next

$$\begin{aligned} \Gamma\left(\frac{1}{2}\right) &= \int_0^{\infty} t^{-\frac{1}{2}} e^{-t} dt \rightarrow [t = x^2 \rightarrow dt = 2x dx] = 2 \int_0^{\infty} x x^{-1} e^{-x^2} dx = \\ &= 2 \int_0^{\infty} e^{-x^2} dx = 2 \frac{\sqrt{\pi}}{2} \rightarrow \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi} \end{aligned}$$

Finally we will use the definition for  $\Gamma(z)$ ,  $\text{Re } z \leq 0$ :

$$\Gamma(z) = \frac{\Gamma(z+n)}{(z+n-1)\dots(z+1)z}, \quad n \text{ chosen such that } \operatorname{Re}(z+n) > 0$$

This gives

$$\Gamma\left(-\frac{1}{2}\right) = \frac{\Gamma\left(-\frac{1}{2}+1\right)}{-\frac{1}{2}} = \frac{\Gamma\left(\frac{1}{2}\right)}{-\frac{1}{2}} \rightarrow \Gamma\left(-\frac{1}{2}\right) = -2\sqrt{\pi}$$

□

## Problem S.2

Show that

$$\sum_{-\infty}^{\infty} |J_n(r)|^2 = 1$$

**Solution:**

Using the Fourier expansion

$$e^{ir \sin \theta} = \sum_{-\infty}^{\infty} J_n(r) e^{in\theta}$$

and Parseval's Theorem

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \overline{e^{ir \sin \theta}} e^{ir \sin \theta} d\theta = \sum_{-\infty}^{\infty} \overline{J_n(r)} J_n(r) e^{in\theta} e^{in\theta}$$

↓

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} d\theta = \sum_{-\infty}^{\infty} |J_n(r)|^2$$

↓

$$\frac{1}{2\pi} 2\pi = \sum_{-\infty}^{\infty} |J_n(r)|^2 = 1$$

□

## Problem S.3

(a) Show by differentiating the power series that

$$\frac{d}{dx} \left( \frac{J_\nu(x)}{x^\nu} \right) = -\frac{J_{\nu+1}(x)}{x^\nu}$$

(b) Use (a) and

$$J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$$

to calculate  $J_{\frac{3}{2}}(x)$

(c) Show that  $J_0'(x) = -J_1(x)$

### Solution:

The power series is

$$\begin{aligned} J_\nu(x) &= \left(\frac{x}{2}\right)^\nu \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4}\right)^k \\ &\quad \downarrow \\ \frac{J_\nu(x)}{x^\nu} &= \left(\frac{1}{2}\right)^\nu \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4}\right)^k \end{aligned}$$

This can be differentiated termwise as

$$\begin{aligned} \frac{d}{dx} \left( \frac{J_\nu(x)}{x^\nu} \right) &= \left(\frac{1}{2}\right)^\nu \sum_{k=0}^{\infty} \frac{-2xk}{4k! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4}\right)^{k-1} = \\ &= -\frac{x}{2} \left(\frac{1}{2}\right)^\nu \sum_{\substack{k=1 \\ *}}^{\infty} \frac{1}{(k-1)! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4}\right)^{k-1} = \\ &= -\frac{1}{x^\nu} \left(\frac{x}{2}\right)^{\nu+1} \sum_{k=1}^{\infty} \frac{1}{(k-1)! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4}\right)^{k-1} \rightarrow [k \rightarrow k+1] \rightarrow \\ &\rightarrow -\frac{1}{x^\nu} \left(\frac{x}{2}\right)^{\nu+1} \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu + 1 + k + 1)} \left(-\frac{x^2}{4}\right)^k = \boxed{-\frac{J_{\nu+1}(x)}{x^\nu}} \end{aligned}$$

\* (The  $k=0$  contribution is zero, since it is a constant and was differentiated.)

(b) Setting  $\nu = \frac{1}{2}$  we get

$$\frac{d}{dx} \left( \frac{J_{\frac{1}{2}}(x)}{x^{\frac{1}{2}}} \right) = -\frac{J_{\frac{3}{2}}(x)}{x^{\frac{1}{2}}}$$

$$\frac{d}{dx} \left( \frac{J_{\frac{1}{2}}(x)}{x^{\frac{1}{2}}} \right) = \frac{d}{dx} \left( \frac{\sqrt{\frac{2}{\pi x}} \sin x}{x^{\frac{1}{2}}} \right) = \frac{d}{dx} \left( \sqrt{\frac{2}{\pi}} \frac{\sin x}{x} \right) = \sqrt{\frac{2}{\pi}} \left( -\frac{1}{x^2} \sin x + \frac{1}{x} \cos x \right)$$

↓

$$J_{\frac{3}{2}}(x) = x^{\frac{1}{2}} \sqrt{\frac{2}{\pi}} \left( \frac{\sin x}{x^2} - \frac{x}{x^2} \cos x \right) = \sqrt{\frac{2}{\pi}} x^{\frac{3}{2}} \left( \frac{\sin x - x \cos x}{x^3} \right)$$

(c) Using the result from (a)

$$\frac{d}{dx} \left( \frac{J_0(x)}{x^0} \right) = -\frac{J_1(x)}{x^0} \rightarrow J_{0'}(x) = -J_1(x)$$

□

## Problem S.4

Determine the general solution to

(a)  $r^2 y'' + r y' = 0$

(b)  $r^2 y'' + r y' - y = 0$

(c)  $r^2 y'' + r y' + r^2 y = 0$

### Solution:

Using a definition that the general solution to Bessel's differential equations

$$u'' + \frac{1}{r} u' + \left( \lambda - \frac{\nu^2}{r^2} \right) u = 0$$

are

$$\begin{aligned} & aJ_{\nu}(\sqrt{\lambda}r) + bY_{\nu}(\sqrt{\lambda}r), & \text{if } \lambda > 0 \\ & ar^{\nu} + br^{-\nu}, & \text{if } \lambda = 0, \nu \neq 0 \\ & a + b \ln r, & \text{if } \lambda = \nu = 0 \end{aligned}$$

(a)

$$r^2 y'' + r y' = 0 \rightarrow y'' + \frac{1}{r} y' = 0, \quad \lambda = \nu = 0$$

$$y(r) = a + b \ln r$$

(b)

$$r^2 y'' + r y' - y = 0 \rightarrow y'' + \frac{1}{r} y' - \frac{1}{r^2} y = 0, \quad \lambda = 0, \nu = 1$$

$$y(r) = ar + \frac{b}{r}$$

(c)

$$r^2 y'' + r y' + r^2 y = 0 \rightarrow y'' + \frac{1}{r} y' + y = 0, \quad \lambda = 1, \nu = 0$$

$$y(r) = aJ_0(r) + bY_0(r)$$

□

### Problem S.5

Let  $I = (0, 2]$

$$\mathcal{A}u = -\frac{1}{x}(xu')'$$

$$D_{\mathcal{A}} = \{u \in \mathcal{C}^2(I) \mid u \text{ bounded near } 0, u'(2) = 0\}$$

- (a) Determine a scalar product in which  $\mathcal{A}$  is symmetric and positive semidefinite
- (b) Determine the eigenvalues and eigenvectors of  $\mathcal{A}$ .
- (c) Determine numerically the three lowest eigenvalues.
- (d) Let  $u(x) = x$ . Write down  $u$ 's expansion in eigenfunctions to  $\mathcal{A}$ . The coefficients should be given in integral form.

### Solution:

(a)

$$(u \mid \mathcal{A}v) = \int_0^2 \overline{u(x)} \mathcal{A}v(x) w(x) dx$$

with  $w(x) = x$

$$\begin{aligned} (u \mid \mathcal{A}v) &= - \int_0^2 \overline{u} (xv')' dx = - \underbrace{[\overline{u}(xv')]_0^2}_{=0} + \int_0^2 (\overline{u}' x) v' dx = \\ &= \underbrace{\int_0^2 (x \overline{u}') v' dx}_{= (*)} = \underbrace{[(x \overline{u}') v]_0^2}_{=0} - \int_0^2 (x \overline{u}')' v dx = (\mathcal{A}u \mid v) \end{aligned}$$

To show that it's positive semidefinite, start from (\*) and set  $u = v$ :

$$\int_0^2 x \overline{u'} u' dx = \int_0^2 x |u'|^2 dx \geq 0$$

(b) Setting

$$\begin{aligned} \mathcal{A}u = \lambda u &\rightarrow -\frac{1}{x}(xu')' = -\frac{1}{x}(u' + xu'') = \lambda u \\ &\downarrow \\ -\frac{1}{x}xu'' - \frac{1}{x}u' - \lambda u &= 0 \rightarrow u'' + \frac{1}{x}u' + \lambda u = 0 \end{aligned}$$

which has the solutions (according to the definition of the Bessel's differential equation solutions):

$$\begin{aligned} aJ_0(\sqrt{\lambda}x) + bY_0(\sqrt{\lambda}x), & \quad \lambda > 0, \nu = 0 \\ a + b \ln r, & \quad \lambda = \nu = 0 \end{aligned}$$

Since  $Y_0$  is unbounded near  $x = 0$ ,  $b$  has to be zero to fulfill the requirement of  $u$  being bounded near zero.

$$\begin{aligned} u(x) &= aJ_0(\sqrt{\lambda}x) \\ u'(2) &= \sqrt{\lambda}aJ_0'(\sqrt{\lambda}2) = 0 \end{aligned}$$

This gives  $\lambda_0 = 0$ . The rest of the eigenvalues are given by

$$J_0'(2\sqrt{\lambda_k}) = 0 \rightarrow \lambda_k = \left(\frac{\beta_{0,k}}{2}\right)^2$$

The eigenfunctions are then given by

$$\varphi_{k(x)} = J_0\left(\frac{\beta_{0,k}}{2}x\right)$$

(c)

$$\begin{aligned} \lambda_0 &= 0 \\ \lambda_1 &= \left(\frac{3.8317}{2}\right)^2 = 3.67 \\ \lambda_2 &= \left(\frac{7.0156}{2}\right)^2 = 12.31 \end{aligned}$$

□